

Multiply $(-13) \times (-20)$ using booth's multiplication.

$$M = (-13)_{10} = (110011)_2$$

$$-M = (13)_{10} = (001101)_2$$

$$Q = (-20)_{10} = (101100)_2$$

$$-Q = (20)_{10} = (010100)_2$$

AC

Q

Q₁

000000	101100	0
000000	010110	0
000000	001011	0
001101	001011	0
000110	100101	1
000011	010010	1
110110	010010	1
111011	001001	0
001000	001001	0
000100	000100	1

28 + 21 = 260

operation.

Q₀Q₁ = 00

ASR

Q₀Q₁ = 00

ASR

Q₀Q₁ = 10

AC = AC - M

ASR

Q₀Q₁ = 11

ASR

Q₀Q₁ = 01

AC = AC + M

ASR

Q₀Q₁ = 10

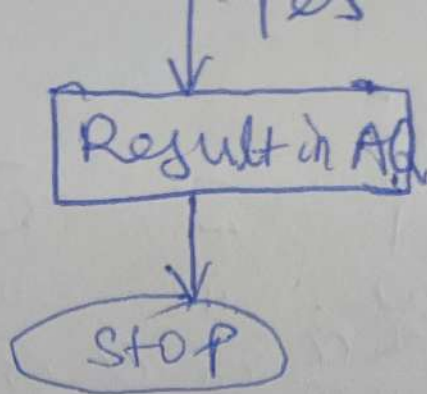
AC = AC - M

ASR

(+)
000000
(-M) 001101
001101

000011
110011
110110

111011
001101
1001000



$$(-7) \times (+3) = (-21)$$

Tracing table

$$M = -7$$

$$-M = (7)_{10} = (0111)_2$$

$$M = (1001)_2$$

n	A	Q	q ₀	Action
4	0000	0011	0	Initialization.
	0111	0011	0	A = A - M.
3	0011	1001	1	ASR
2	0001	1100	1	ASR
	1010	1100	1	A = A + M
1	1101	0110	0	ASR
0	1110	1011	0	ASR

$$\begin{array}{r}
 0000 \\
 + 0111 \\
 \hline
 0111
 \end{array}$$

$$\begin{array}{r}
 0001 \\
 1001 \\
 \hline
 1010
 \end{array}$$

If MSB bit is 1, result is (-)ve & ~~2's complement form~~.

→ If MSB 1, then find its 2's comp. for getting magnitude.

→ If MSB 0, directly find decimal equivalent.